Hierarchical Frequency–Spatial Integration Network for Remote Sensing Super-Resolution

Yunong Chen, Zhenfeng Shao, Josue Antonio Nescolarde Selva, John van Genderen, and Deren Li

***Abstract*—Remote sensing image (RSI) super-resolution (SR) offers a cost-effective and efficient approach to obtain high-resolution imagery, playing a crucial role in enhancing the precision of image analysis. However, existing methods suffer from an inherent trade-off between performance and efficiency, and their over-reliance on low-level features leads to insufficient fusion of multiscale information, low detail fidelity in reconstructed results, along with overfitting and artifacts. To address these issues, this article proposes a novel hierarchical frequency–spatial interaction network (HiFi-Net) for RSI SR. The proposed HiFi-Net incorporates a frequency–spatial attention module that leverages Fourier transformation to efficiently model global dependencies, thereby enhancing perceptual scope while reducing computational complexity. Furthermore, HiFi-Net employs a hierarchical interaction mechanism, comprised of a hierarchical spatial cross-attention (HSCA) module and a hierarchical channel cross-attention module, to facilitate cross-level feature modulation. This design effectively integrates low-level high-frequency details with high-level features, thereby adaptively aggregating critical information and resulting in superior SR performance. Extensive experiments on several public remote sensing datasets demonstrate that HiFi-Net consistently outperforms state-of-the-art methods in both quantitative metrics and visual quality.**

***Index Terms*—Fourier transform, frequency–spatial attention, hierarchical cross-attention, remote sensing image (RSI), super-resolution (SR).**

I. Introduction

HIGH-RESOLUTION (HR) remote sensing imagery provides richer textual details and structural information than its low-resolution (LR) counterpart, making it indispensable for fine-grained visual tasks [1]. In recent decades, rapid advances in imaging technology have substantially improved the spatial resolution of remote sensing images (RSIs). Nevertheless, growing application demands across various domains (e.g., object detection [2], medical imaging [3], mapping [4], fine-grained classification [5], and change monitoring [6], [7]) continue to drive the need for imagery with even higher spatial resolution. This is particularly evident in remote sensing, where HR images play a crucial role in supporting accurate analysis and informed decision making [8], [9], [10]. Super-resolution (SR) techniques, which offer a viable pathway to enhance image resolution, have emerged as a major research focus.

Formally, the degradation process of the HR image $\mathbf{I_{HR}}$ can be defined as follows:

$$\mathbf{I_{LR}} = \Psi_{\text{down}_s}(\mathbf{I_{HR}} \circledast \mathbf{k}) + \boldsymbol{\beta} \tag{1}$$

where $\mathbf{I_{HR}} \in \mathbb{R}^{H\times W\times C}$ denotes observed LR image and $\mathbf{I_{HR}} \in \mathbb{R}^{sH\times sW\times C}$ is the corresponding HR image, where H and W are the height and width, C is the number of feature channels, s is the scale factor, $\mathbf{k}$ represents the blur kernel, $\circledast$ indicates the 2-D convolution operation, Ψ_{down_s} stands for the downsampling function with scale factor s, and $\boldsymbol{\beta}$ models the additive noise component.

The SR task is fundamentally an ill-posed inverse problem, which can be formulated as the HR image recovering from LR inputs. To generate a unique and stable solution, it is a common practice to introduce natural image priors or constraints [11]. Thus, various SR methods are proposed to address the ill-posedness. In general, SR methods can be broadly categorized into three main categories: interpolation-based SR [12], reconstruction-based SR [13] methods, and learning-based SR method [14], [15], [16].

Specifically, interpolation-based SR methods, such as linear or bicubic interpolation, generate new pixel values by extrapolating from known data points within a local neighborhood. However, such methods often fail to reconstruct high-frequency details effectively, resulting in overly smooth outputs with blurred edges and textures [17]. Reconstruction-based SR methods, on the other hand, incorporate explicit prior knowledge or regularization constraints, such as local smoothness [18], self-similarity [19], or sparsity [20], to guide the inference of high-resolution images. A major limitation of reconstruction-based SR methods is the reliance on hand-crafted priors, which may not generalize well across diverse scenes and target types. When the input images deviate from the assumed prior model, the reconstruction performance tends to degrade significantly [21].

With the remarkable advances in artificial intelligence technologies and the exponential growth of satellite imagery over recent decades, learning-based SR methods have demonstrated

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strong capabilities in modeling degradation patterns, particularly when trained on large-scale datasets [22]. Among these, deep-learning (DL)-based SR methods have emerged as the dominant paradigm, with pioneering works such as SRCNN [20] laying the foundation. SRCNN introduced the first convolutional-neural-network (CNN)-based SR framework, employing a simple three-layer architecture that nonetheless outperformed traditional bicubic interpolation. However, its computational cost remained prohibitive due to processing operations being performed in the upsampled HR space. To address this limitation and capture more complex LR-to-HR mappings, subsequent research has focused on developing deeper, more sophisticated network architectures that perform most operations in the low-resolution space, thereby significantly reducing computational cost. This efficiency gain enables the use of deeper networks that can learn more expressive end-to-end mappings from LR to HR without relying on handcrafted priors [23], [24], [25], [26].

However, the majority of the above learning-based SR methods have been primarily developed and optimized for natural images. When applied to the remote sensing domain, the distinct characteristics of RSI must be considered [27]. Compared with natural images, remote sensing data exhibit several unique characteristics: highly diverse backgrounds due to varied terrain types, multiscale objects resulting from vastly different ground target sizes, rich spectral information, and frequent atmospheric interference such as clouds and haze [28]. Due to the intrinsic complexity of RSI and various physical–technical constraints, RSI SR often suffers from severe performance degradation. Reconstructed RSI typically exhibits notably lower spatial resolution and inferior visual fidelity compared to natural images [28]. This pronounced performance gap underscores the necessity for specialized RSI SR solutions.

In the early development of RSI SR, a number of CNNs are introduced, with methods such as LGCNet [29] and DCM [30] focusing on foundational designs. To further enhance performance, subsequent research has incorporated more advanced architectural innovations, as seen in SRDSRAN [14], HSENet [19], FeNet [31], HATNet [32], and so on [16], [33], [34]. For example, Lei proposed HSENet [19], which achieves favorable SR results by exploiting hybrid-scale self-similarity. However, the current SR method of RSI remains constrained by two key issues.

1) *Inherent Trade-Off Between Performance and Efficiency:* In existing RSI SR networks, local convolutional networks excel at capturing detailed features but are limited by their receptive fields; global attention models can model long-range dependencies yet incur high computational complexity. Constraints imposed by computational complexity further result in an inherent trade-off between performance and efficiency for SR models.
2) *Over-Reliance on Low-Level Features:* In handling high-repeatability features across different scales and locations in RSIs, existing SR methods universally rely solely on low-level features, which only fails to effectively fuse multilevel information for high-fidelity detail reconstruction but also results in issues such as low detail fidelity, potential overfitting, and artifacts in the reconstructed outputs.

A. Main Works

To address the aforementioned issues, this article proposes a novel Hierarchical Frequency–Spatial Interaction Network, called HiFi-Net, for RSI SR. Specifically, HiFi-Net first extracts low-level features of RSIs via cascaded convolutions, then employs customized feature-enhanced frequency–spatial attention (FFA) modules to capture cross-domain (frequency/spatial) features of RSIs, as this design efficiently models global dependencies while reducing computational complexity. Afterward, these enhanced features are fused with the initial low-level features through the hierarchical spatial cross-attention (HSCA) module. In the reconstruction phase, the upsampled high- and low-level features are processed by the hierarchical channel cross-attention (HCCA) module, enabling high-level features to modulate low-level details and achieve cross-level feature information complementarity. This dual innovative design (frequency–spatial attention and hierarchical interaction) realizes adaptive aggregation of critical information in RSIs, ultimately delivering superior SR reconstruction performance. The main contributions of this work are summarized as follows.

1) We propose HiFi-Net, a novel RSI SR framework that integrates a frequency–spatial attention mechanism with hierarchical feature interaction to achieve superior reconstruction performance.
2) We design an FFA module for HiFi-Net that conducts attention computation via Fourier transform, which not only reduces the computational complexity of spatial-domain operations but also strengthens global modeling capability through efficient cross-feature interaction.
3) We introduce a hierarchical interaction mechanism for HiFi-Net, which is designed to strategically aggregate multilevel features to reconstruct fine spatial details while enriching channelwise representations, thereby addressing the limitations of insufficient multiscale fusion in existing methods.

Extensive experiments on several public RSI datasets demonstrate that our proposed HiFi-Net achieves state-of-the-art (SOTA) performance, producing visually superior images with suppressed artifacts. Moreover, comprehensive ablation studies further confirm the efficacy of the core modules proposed in HiFi-Net.

B. Organizations

The remainder of this article is structured as follows: Section II reviews related works on natural image and RSI SR; Section III details the methodology and architectural design of our proposed HiFi-Net; Section IV presents experimental setups and comprehensive comparisons between HiFi-Net and other SOTA methods; Section V presents the ablation studies; and Section VI concludes with a summary of our key findings.

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II. Related Works

The field of SR has evolved from early interpolation methods to deep learning, progressing from improving accuracy to optimizing efficiency and developing domain-specific solutions.

A. Natural Image SR Methods

The pioneering work of SRCNN [20] demonstrated for the first time the effectiveness of an end-to-end CNN in learning the mapping from LR to HR images, laying the foundation for DL-based SR methods. However, SRCNN suffered from computational inefficiency. To address this, FSRCNN [35] introduced critical improvements by proposing a compact, hourglass-shaped CNN architecture. It reduced the feature dimension before mapping, employed smaller but deeper filters, and incorporated a deconvolution layer at the network end for direct upsampling from the original LR image, significantly accelerating the model while maintaining performance of nature image SR. Moreover, VDSR [36] constructed a very deep network and facilitated training convergence by learning residuals and employing extremely high learning rates, proving that greater depth leads to substantial gains in natural image reconstruction accuracy.

A significant advancement of nature image SR came with the integration of attention mechanisms, which greatly enhance the representational power of CNNs [15], [37], [38]. Parallelly, the exploration of frequency-domain analysis has provided another dimension for improvement. The Fourier transform has been leveraged to capture long-range dependencies and expand the receptive field [39]. However, a common limitation among the frequency-domain nature image SR methods is that they primarily use Fourier convolutions to process complex frequency features, often overlooking the potential of amplitude-phase separation to disentangle global common image structures from specific degradations. Subsequently, the research focus shifted toward increasing network depth and diversity to boost SR performance, such as IDM [40], GaussianSR [41], and DiT4SR [42].

In contrast to general image SR, the task is more challenging for RSIs due to their inherent imaging complexity and constraints, often leading to severe performance degradation.

B. RSI SR Methods

Extending SR to the remote sensing domain presents distinct challenges beyond those encountered in natural image SR, necessitating explicit consideration of imagery-specific attributes like complex textures and multiscale structures. Consequently, tailored methodologies have been proposed, such as LGCNet [29], HSENet [19], and FeNet [31]. Specifically, LGCNet [29] introduced a multifork structure to capture multilevel representations that encompass both local details and global environmental priors, thereby adapting to the complexity of remote sensing scenes. However, its reliance on convolutional operations limits its capability to model long-range dependencies. HSENet [19] leveraged the inherent internal self-similarity of RSIs through specially designed modules to learn more powerful feature representations. To meet the computational constraints of practical deployment, FeNet [31] adopted a lightweight design, employing a lattice block and nested feature enhancement blocks to significantly reduce parameters while maintaining performance.

To address the limitation of most RSI SR models that perform upsampling only at the final stage, TransENet [27] introduced a Transformer-based multistage enhancement structure to effectively fuse features after upsampling, thereby overcoming previous constraints in high-dimensional feature extraction. Nevertheless, Transformer-based methods cannot fully exploit self-attention due to limited receptive fields. Accordingly, HATNet [43] developed a hybrid attention Transformer for RSI SR that integrates channel attention with window-based self-attention and incorporates an overlapping cross-attention module to enhance cross-window interaction. This design activates more input pixels for reconstruction and, when combined with same-task pretraining, achieves significant performance gains [44].

While HAUNet [32] effectively mitigated spectral distortion through hybrid attention mechanisms, its architecture shows limited capability in preserving complex structural patterns under large scaling factors, especially in reconstructing intricate geometric details. Similarly, DCM [30] harmonized various architectural components, such as residual and skip connections and parallel convolutions, and alleviates degradation in deep networks, yet its relatively fixed feature fusion strategy restricts its adaptability to diverse remote sensing scenarios. Meanwhile, the GAN-based method, named SRDSRAN [14], employed adversarial training to achieve high-resolution reconstruction under large scaling factors (e.g., $\times 8$).

Recent advanced RSI SR methods have shifted toward task-driven and fusion-aware frameworks. For example, JointSeg [45] directly generates 1-m resolution impervious surface area maps from 10-m resolution Sentinel-2 imagery, eliminating the dependence on expensive commercial very-high-resolution images. CasFormer [46] proposed a cascaded Transformer architecture specifically for fusion-aware hyperspectral image reconstruction in dual-camera coded aperture snapshot spectral imaging systems. Unlike conventional state space models, MambaX [47] employed customized neural modules to adaptively compute control parameters, enabling more flexible approximation of complex degradation processes. Moreover, PLAN [48] constructed a parallel lattice attention network to enhance the reconstruction capability of complex features in SR.

In brief, the integration of convolutional architectures (e.g., LGCNet [29], HSENet [19], and FeNet [31]), generative adversarial networks (e.g., SRDSRAN [14]), state space models (e.g., MambaX [47]), and Transformer-based attention mechanisms (e.g., HATNet [43] and TransENet [27]) reflects sustained efforts to address key challenges in RSI SR tasks. However, existing methods are still limited by the inherent trade-off between performance and efficiency, and rely excessively on low-level features, resulting in insufficient multiscale information fusion, reduced detail fidelity of reconstructed results, as well as problems such as overfitting and artifacts. Nevertheless, it remains a significant challenge to develop

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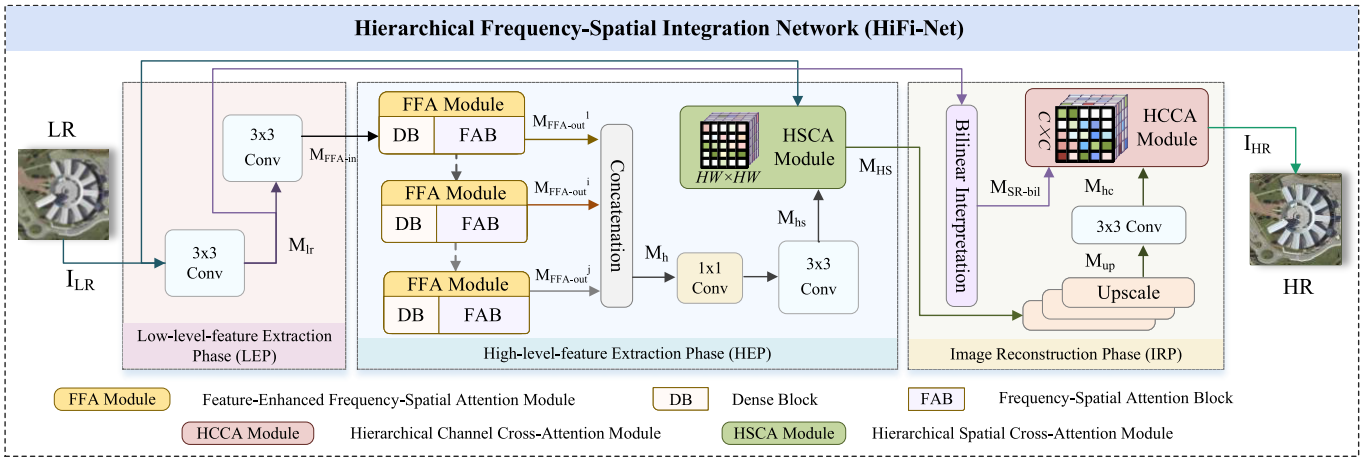


Fig. 1. Overall architecture of the proposed HiFi-Net, which consists of three phases (LEP, HEP, and IPR) and three main modules (FFA, HSCA, and HCCA).

intelligent RSI SR methods capable of recovering fine-grained textures and spectral characteristics under real-world degradation conditions.

III. Methodology

HiFi-Net is an SR network tailored for RSIs, aimed at reconstructing high-resolution outputs with sharp details and authentic fidelity. As illustrated in Fig. 1, the overall framework of HiFi-Net follows a three-phase sequential pipeline: low-level-feature extraction phase (LEP), high-level-feature extraction phase (HEP), and image reconstruction phase (IRP). The SR process begins with the LEP, which employs convolutional operations to extract low-level features and adjust the spatial dimensions for subsequent phases. Next, the HEP performs high-level feature refinement through its two core modules: FFA module and HSCA module. Finally, the IRP utilizes HCCA module to synthesize the final HR image.

A. Model Formulation

1) Low-Level-Feature Extraction Phase: The LEP employs two 3×3 convolutional layers, $\mathcal{G}_{LE}(\cdot)$ represents the mapping function of the LEP that processes the LR input image $\mathbf{I_{LR}}$ to produce the low-level feature representation $\mathbf{M}_{lr} \in \mathbb{R}^{H \times W \times C}$. This process can be formulated as follows:

$$\mathbf{M}_{lr} = \mathcal{G}_{LE}\left(\mathbf{I}_{LR}\right). \tag{2}$$

2) High-Level-Feature Extraction Phase: Following the LEP, the HEP performs high-level feature extraction. This transformation is mainly accomplished by N FFAs and an HSCA. Specifically, the FFA module performs global attention computation via Fourier transform, while the HSCA reconstructs spatial details by extracting and integrating multilevel features. The ith ($i = 1, 2, \ldots, N$) FFA can be formulated as follows:

$$\mathbf{M}_{FFA-out}^{i} = \mathcal{G}_{FFA}^{i}\left(\mathbf{M}_{FFA-in}^{i}\right), \quad i = 1, 2, \ldots, N \tag{3}$$

where N represents the number of FFA modules, $\mathbf{M}_{FFA-out}^{i}$ denotes the output feature maps of the ith FFA module, and $\mathcal{G}_{FFA}^{i}(\cdot)$ represents the function of the ith FFA module. Then, the outputs of all N FFA modules are concatenated along the channel dimension for progressive refinement, yielding the intermediate high-level feature maps $\mathbf{M}_h$.

Subsequently, the HSCA module performs multiscale feature fusion. As shown in Fig. 1, HSCA module takes two inputs: $\mathbf{M}_{lr}$ from the LEP and $\mathbf{M}_{hs}$ from the FFA stack and convolutional layers. Through a cross-attention mechanism, HSCA enables the high-level features to guide and modulate the spatial details in the low-level features, effectively integrating complementary information across hierarchies. The operation of the HSCA module is defined as follows:

$$\mathbf{M}_{HS} = \mathcal{G}_{HSCA}\left(\mathbf{M}_{lr}, \mathbf{M}_{hs}\right) \tag{4}$$

where $\mathcal{G}_{HSCA}(\cdot)$ denotes the function of the HSCA module, and $\mathbf{M}_{HS}$ denotes the final, refined high-level feature representation produced by HSCA module. Finally, the overall procedure of HEP can be defined as follows:

$$\mathbf{M}_{HS} = \mathcal{G}_{HE}\left(\mathbf{M}_{lr}\right) \tag{5}$$

where $\mathcal{G}_{HE}(\cdot)$ represents the composite mapping function, and $\mathbf{M}_{HS} \in \mathbb{R}^{H \times W \times C}$ represents the high-level feature representations.

3) Image Reconstruction Phase: The IRP reconstructs the final HR image $\mathbf{I}_{HR}$ by integrating upsampling, bilinear interpolation, and the HCCA module. Let $\mathcal{G}_{IRP}(\cdot)$ denote the composite reconstruction function of this phase, which can be formulated as follows:

$$\mathbf{I}_{HR} = \mathcal{G}_{IRP}\left(\mathbf{M}_{HS}\right). \tag{6}$$

The reconstruction begins with a subpixel convolution for upsampling. First, a convolutional layer extracts features and expands the channel dimension of $\mathbf{M}_{HS}$ from D to s^2D, where s is the scaling factor. A pixel-shuffle layer then rearranges the expanded features, restoring the channel dimension to D while increasing the spatial resolution to $sH \times sW$, yielding the upscaled feature maps $\mathbf{M}_{up}$. Subsequently, another convolutional layer is employed to reduce the channel dimension of $\mathbf{M}_{up}$ (D) and generate three-channel dimension feature maps $\mathbf{M}_{hc}$. In parallel, bilinear interpolation is employed to

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upsample the low-level feature maps $\mathbf{M}_{lr}$ from the LEP to the target resolution, generating $\mathbf{M}_{SR-bil} \in \mathbb{R}^{sH \times sW \times 3}$.

Finally, the HCCA module further enhances detail recovery by aligning and refining $\mathbf{M}_{hc}$ with $\mathbf{M}_{SR-bil}$ through channel-wise cross-attention, producing the final HR image $\mathbf{I}_{HR}$

$$\mathbf{I}_{HR} = \mathcal{G}_{HCCA}\left(\mathbf{M}_{hc}, \mathbf{M}_{SR-bil}\right). \tag{7}$$

This integrated design ensures both high-level information and low-level spatial details are preserved and enhanced.

4) Loss Function: The proposed HiFi-Net is optimized end-to-end using the ℓ_1-based loss function, which promotes sharp edge reconstruction and rich texture recovery by mitigating the oversmoothing tendency, thus is well-suited for the RSI SR tasks. Given a training dataset $\{\mathbf{I}^{\eta}_{\mathbf{LR}}, \mathbf{I}^{\eta}_{\mathbf{HR}}\}_{\eta=1}^{\mathcal{Z}}$ containing $\mathcal{Z}$ LR and HR pairs, the loss function of HiFi-Net is defined as follows:

$$\mathcal{L}(\xi) = \frac{1}{\mathcal{Z}} \sum_{\eta=1}^{\mathcal{Z}} \left\| \mathbf{I}^{\eta}_{LR-HiFi} - \mathbf{I}^{\eta}_{HR-GT} \right\|_1 \tag{8}$$

where ξ denotes all learnable parameters of the network, $\mathcal{Z}$ is the total number of training images, $\mathbf{I}^{\eta}_{LR-HiFi}$ is the SR output of the ith LR input, and $\mathbf{I}^{\eta}_{HR-GT}$ is the corresponding ground-truth (GT) HR image.

B. Feature-Enhanced Frequency–Spatial Attention

Traditional convolutional layers are limited by their local receptive fields, making it difficult to model long-range dependencies. While attention mechanisms can overcome this limitation and capture global context, their computational complexity in the spatial domain, requiring the calculation of pairwise associations between all pixels, becomes prohibitively high for large inputs. The Fourier transform offers an effective solution to this problem by converting signals from the spatial domain to the frequency domain. In this domain, global dependencies can be approximated through efficient elementwise operations, bypassing the need for extensive matrix multiplications and significantly improving computational efficiency.

Building on this insight, our proposed FFA module first employs convolution to extract foundational features and subsequently integrates the Fourier Transform with attention mechanisms for efficient global feature integration. As illustrated in Fig. 2, each FFA module is composed of two fundamental building blocks: the dense block (DB) and the frequency–spatial attention block (FAB). Within the DB, features from all preceding layers are integrated via channelwise concatenation. This design promotes feature reuse and enriches representations while maintaining a compact parameter space. The concatenated feature maps $\mathbf{M}_{DB-out}$ are then fused by a 1×1 convolutional layer to generate $\mathbf{M}_{FFA-in}$, which serves as the input to the FAB. The DB enriches local feature representation and provides the FAB with a comprehensive feature set for subsequent global context modeling. Then, FAB is applied after the DB to perform global optimization on these already enriched features. This design avoids the inefficiency that would arise from modeling global dependencies on less informative low-level features.

The FAB transforms the feature maps $\mathbf{M}_{FFA-in}$ into the frequency domain via a fast Fourier transform (FFT). In this

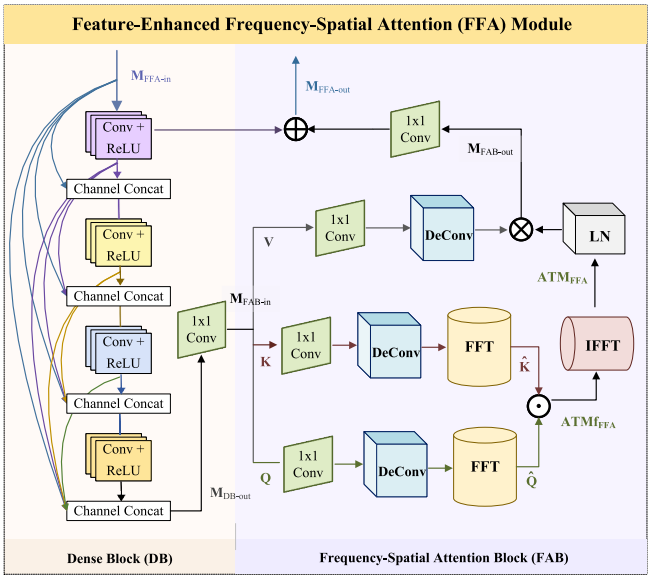


Fig. 2. Our customized FFA module in HiFi-Net, which composes DB and FAB. DB enhances feature representation capacity, and FAB efficiently captures global dependencies.

domain, an attention mechanism performs weighting on the frequency components to enhance informative structures, after which an inverse FFT maps the features back to the spatial domain. This domain transition facilitates efficient modeling of global context and long-range dependencies. We leverage the convolution theorem, which states that spatial convolution is equivalent to elementwise multiplication in the frequency domain. Based on this property, we replace the computationally intensive spatial attention operation with efficient frequency-domain computation. Furthermore, we leverage the convolution theorem, which states that spatial convolution is equivalent to elementwise multiplication in the frequency domain. Based on this property, we replace the computationally intensive spatial attention operation (i.e., the $\mathbf{K^T Q}$ matrix multiplication) with efficient frequency-domain computation to generate the attention map.

Note that the input feature maps to the ith FFA module, denoted as $\mathbf{M}^{i}_{FFA-in}$, are taken as the output from the preceding $(i-1)$th module, i.e., $\mathbf{M}^{i}_{FFA-in} = \mathbf{M}^{i-1}_{FFA-out}$. This sequential connection ensures continuous feature propagation. The internal process of the ith FFA module is detailed as follows. First, the input features $\mathbf{M}^{i}_{FFA-in}$ are projected via composite convolutions into query ($\mathbf{Q}_i$), key ($\mathbf{K}_i$), and value ($\mathbf{V}_i$) representations, all of which lie in $\mathbb{R}^{H \times W \times C}$. The FFT is then applied to $\mathbf{Q}_i$ and $\mathbf{K}_i$, transforming them into the frequency domain to obtain $\hat{\mathbf{Q}}_i$ and $\hat{\mathbf{K}}_i$. The attention map $\mathbf{ATMf}^{i}_{FFA}$ is computed in the frequency domain via the elementwise product of $\hat{\mathbf{Q}}_i$ and $\hat{\mathbf{K}}_i$. Subsequently, the inverse fast Fourier transform (IFFT) is used to convert $\mathbf{ATMf}^{i}_{FFA}$ back to the spatial domain, yielding the spatial attention map $\mathbf{ATM}^{i}_{FFA}$. This attention map is then normalized using layer normalization (LN) and matrix multiplied with the value feature $\mathbf{V}_i$. Finally, the result is passed through a 1×1 convolution and added to the input feature $\mathbf{M}^{i}_{FFA-in}$ via a residual connection, producing the final output $\mathbf{M}^{i}_{FFA-out}$ of this FFA module. Mathematically, this

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process can be formulated as follows:

$$\{\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i\} = \Psi_{Conv}^{3\times3}\left(\Psi_{Conv}^{1\times1}\left(\mathbf{M}_{FFA-in}^{i}\right)\right) \tag{9}$$

$$\{\hat{\mathbf{Q}}_i, \hat{\mathbf{K}}_i\} = \Psi_{\text{FFT}}\left(\mathbf{Q}_i\right), \Psi_{\text{FFT}}\left(\mathbf{K}_i\right) \tag{10}$$

$$\mathbf{ATMf}_{FFA}^{i} = \hat{\mathbf{Q}}_i \odot \hat{\mathbf{K}}_i \tag{11}$$

$$\mathbf{ATM}_{FFA}^{i} = \Psi_{\text{FFT}^{-1}}\left(\mathbf{ATMf}_{FFA}^{i}\right) \tag{12}$$

$$\mathbf{M}_{FFA-out}^{i} = \Psi_{\text{LN}}\left(\mathbf{ATM}_{FFA}^{i}\right)\mathbf{V}_i + \mathbf{M}_{FFA-in}^{i} \tag{13}$$

where $\Psi_{Conv}^{3\times3}$ denotes the 3×3 depthwise convolution, $\Psi_{Conv}^{1\times1}$ denotes the 1×1 pointwise convolution, Ψ_{FFT} and Ψ_{FFT}^{-1} represent the FFT and IFFT, respectively, Ψ_{LN} denotes the LN operation applied to the spatial attention map for stabilization, and $\odot$ signifies elementwise multiplication. Specifically, computing $\hat{\mathbf{Q}}_i \odot \hat{\mathbf{K}}i$ in the frequency domain is equivalent to learning a global convolution kernel that, when applied to the spatial features, captures long-range dependencies across the entire image. Unlike spatial attention which computes pairwise similarities between all pixels (complexity $O(N^2)$ with $N = H \times W$), frequency-domain attention achieves global modeling with $O(N \log N)$ complexity via FFT. The learned $\mathbf{ATM}_{FFA}^{i}$ after IFFT can be interpreted as a spatial attention map that highlights regions requiring global context, but it is generated through frequency-domain filtering rather than explicit pairwise comparisons.

A critical hyperparameter in the dense connectivity is the channel dimension (or growth rate) D, which defines the dimension of output channels produced by each convolutional layer within the block. Specifically, the output feature map $\mathbf{M}_{\text{FFA}}^{i,j}$ of the jth convolutional layer in the ith FFA module has D channels. Consequently, after concatenating features from all preceding layers, the input to the jth convolutional layer exhibits a channel dimension of $D_0 + (j-1)D$, where D_0 denotes the channel dimension of the initial input $\mathbf{M}_{\text{FFA-out}}^{i-1}$ to the block. This linear growth in channel dimension enables the network to progressively accumulate and refine feature representations. The process for generating $\mathbf{M}_{\text{FFA}}^{i,j}$ is formally expressed as follows:

$$\mathbf{M}_{\text{FFA}}^{i,j} = \Psi_{\text{ReLU}}\left(\Psi_{\text{Conv}}^{3\times3}\left(\Psi_{\text{Concat}}\left(\mathbf{M}_{\text{FFA-out}}^{i-1}, \mathbf{M}_{\text{FFA}}^{i,1}, \times \mathbf{M}_{\text{FFA}}^{i,2}, \mathbf{M}_{\text{FFA}}^{i,3}, \ldots, \mathbf{M}_{\text{FFA}}^{i,j-1}\right)\right)\right) \tag{14}$$

where $\mathbf{M}_{\text{FFA-out}}^{i-1}$ denotes the output of the $(i-1)$th FFA module, serving as the input to the current module, and $\Psi_{\text{Concat}}(\mathbf{M}_{\text{FFA-out}}^{i-1}, \mathbf{M}_{\text{FFA}}^{i,1}, \mathbf{M}_{\text{FFA}}^{i,2}, \ldots, \mathbf{M}_{\text{FFA}}^{i,j-1})$ concatenates the output feature map from the 1st to the $(j-1)$th convolutional layers within $(i-1)$th FAB and current ith FAB. The concatenation of all feature maps along the channel dimension ensures that subsequent layers can fully leverage the multiscale features extracted by preceding layers.

C. Hierarchical Cross-Attention Modules

Different from conventional multiscale fusion SR approaches, we propose the HSCA and HCCA modules by computing attention in the spatial and channel dimensions respectively, enabling fully leverage coarse-grained features from low levels and fine-grained features from high levels to hierarchically integrate key information.

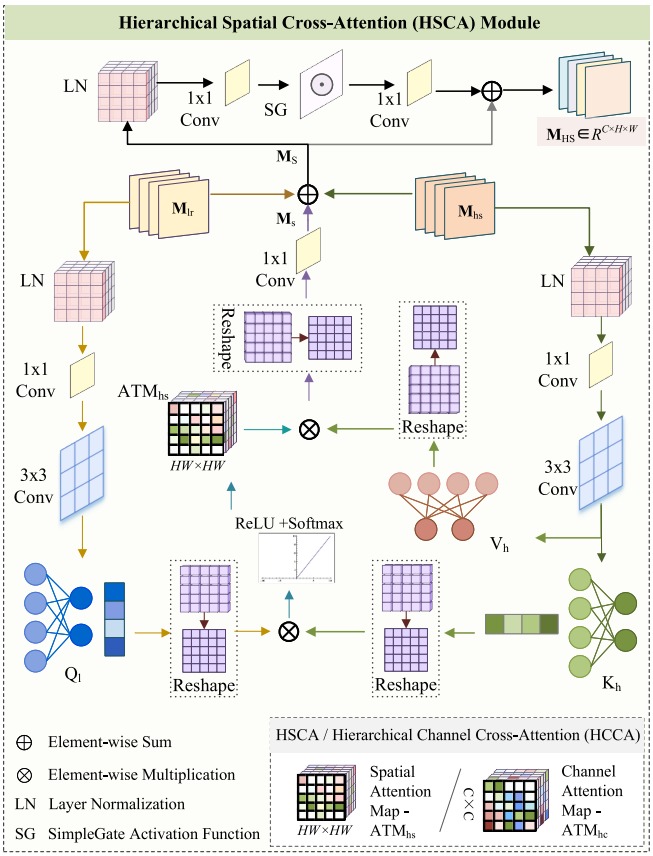


Fig. 3. HSCA module. The query input $\mathbf{Q}_l$ derived from the coarse-grained features $\mathbf{M}_{lr}$ is indicated by the yellow information flow, while the key $\mathbf{K}_h$ and value $\mathbf{V}_h$ obtained from the fine-grained features $\mathbf{M}_{hs}$ are indicated by the green information flow.

1) HSCA Module: Fig. 3 illustrates the structure of the HSCA module. The coarse-grained feature map $\mathbf{M}_{lr}$ and the fine-grained feature map $\mathbf{M}_{hs}$ are first processed to generate the $\mathbf{Q}_l$, $\mathbf{K}_h$, and $\mathbf{V}_h$ projections. Specifically, both feature maps are independently normalized by LN for stability and then passed through a 1×1 convolution for channel adjustment, followed by a 3×3 convolution for spatial feature extraction. This process is formulated as follows:

$$\boldsymbol{Q}_l = \Psi_{\text{Conv}}^{3\times3}\left(\Psi_{\text{Conv}}^{1\times1}\left(\Psi_{\text{LN}}\left(\mathbf{M}_{lr}\right)\right)\right) \tag{15}$$

$$\{\mathbf{K}_h, \mathbf{V}_h\} = \Psi_{\text{Conv}}^{3\times3}\left(\Psi_{\text{Conv}}^{1\times1}\left(\Psi_{\text{LN}}\left(\mathbf{M}_{hs}\right)\right)\right). \tag{16}$$

Subsequently, the tensors $\mathbf{Q}_l$, $\mathbf{K}_h$, and $\mathbf{V}_h$ are reshaped into token $(\tilde{\mathbf{Q}}_l, \tilde{\boldsymbol{K}}_h, \tilde{\boldsymbol{V}}_h)$ using a reshape operation $\mathcal{R}(\cdot)$. The ReLU function (Ψ_{ReLU}) and the Softmax function (Ψ_{Softmax}) are then applied to the result of the matrix multiplication to compute the cross-attention map $\mathbf{ATM}_{hs}$

$$\mathbf{ATM}_{hs} = \Psi_{\text{Softmax}}\left(\Psi_{\text{ReLU}}\left(\tilde{\mathbf{K}}_h^{\top}\tilde{\mathbf{Q}}_l/t\right)\right) \tag{17}$$

where $\tilde{\mathbf{K}}_h^{\top}$ is the transpose matrix of $\tilde{\mathbf{K}}_h$, and t is a temperature coefficient that stabilizes gradient magnitudes during training. This design enables the high-level features to selectively refine the spatial details in the low-level features through a focused, tokenwise interaction.

The attention map $\mathbf{ATM}_{hs}$ is then multiplied with $\tilde{\mathbf{V}}_h$. The result is reshaped back to $\mathbb{R}^{C\times H\times W}$ to obtain the refined feature

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map $\mathbf{M}_s$. Thereafter, a residual aggregation is performed by elementwise summing the coarse-grained feature map $\mathbf{M}_{lr}$, fine-grained feature map $\mathbf{M}_{hs}$, and the newly refined feature map $\mathbf{M}_s$, yielding an intermediate feature representation $\mathbf{M}_S$. Specifically, $\mathbf{M}_S$ is first normalized by LN, then passes through two 1×1 convolutional layers, with a SimpleGate nonlinear activation function (SG(·)) inserted between them. The SimpleGate function splits the feature map along the channel dimension into two parts and performs elementwise multiplication, effectively modeling nonlinear interactions while reducing channel redundancy. Finally, the processed features are combined with the original $\mathbf{M}_s$ via a residual connection to produce the final output $\mathbf{M}_{HS}$ of the HSCA module. This process is defined as follows:

$$\mathbf{M}_S = \mathbf{M}_{lr} + \mathbf{M}_{hs} + \mathbf{M}_s \tag{18}$$

$$\mathbf{M}_{HS} = \Psi_{\text{Conv}}^{1\times1}\left(\Psi_{\text{SG}}\left(\Psi_{\text{Conv}}^{1\times1}\left(\Psi_{\text{LN}}\left(\mathbf{M}_S\right)\right)\right)\right) + \mathbf{M}_S. \tag{19}$$

2) Hierarchical Channel Cross-Attention Module: The HCCA module follows a similar processing pipeline to the HSCA module but exhibits a key distinction in their respective attention maps: HCCA produces a $C \times C$ channel attention map, whereas HSCA generates a $HW \times HW$ spatial attention map. This difference stems from how the input tensors $\mathbf{Q}_l, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{C\times H\times W}$ are reshaped. In the HCCA module, the channel dimension C is treated as the primary dimension, resulting in $\tilde{\mathbf{Q}}_l \in \mathbb{R}^{HW\times C}$, $\tilde{\mathbf{K}}_h \in \mathbb{R}^{C\times HW}$, and $\tilde{\mathbf{V}}_h \in \mathbb{R}^{C\times HW}$. Correspondingly, in the HSCA module, the spatial dimension HW is prioritized, yielding $\tilde{\mathbf{Q}}_l \in \mathbb{R}^{C\times HW}$, $\tilde{\mathbf{K}}_h \in \mathbb{R}^{HW\times C}$, and $\tilde{\mathbf{V}}_h \in \mathbb{R}^{HW\times C}$.

Furthermore, unlike traditional self-attention- based modules that compute correlations within a single feature map, our proposed HSCA and HCCA modules facilitate interactive fusion between features from different levels and granularities. HSCA and HCCA modules then integrate coarse-grained features, which provide fundamental structure and edge information, with the fine-grained features that contribute rich textures and object details.

Consequently, the HSCA module is position-sensitive and excels at capturing spatial details, such as the specific locations and shapes of objects, thereby effectively modeling local structures and fine-grained textures. In contrast, the HCCA module performs channelwise recalibration, selectively emphasizing informative channels to suppress redundancy and thereby improves the generalization capability of the HiFi-Net.

D. Image Reconstruction

The image reconstruction is designed to align and integrate complementary information from multiple feature levels to produce the final HR output. This process is formulated as follows:

$$\mathbf{I}_{\text{HR}} = \mathcal{G}_{HCCA}\left(\Psi_{\text{Bil}}\left(\mathbf{M}_{SR-bil}\right), \mathbf{M}_{\text{hc}}\right). \tag{20}$$

As shown in Fig. 1, it takes $\mathbf{M}_{lr}$ and $\mathbf{M}_{HS}$ as inputs. For $\mathbf{M}_{lr}$, a parameter-free bilinear interpolation operation, $\Psi_{\text{Bil}}(\cdot)$ is applied to upsample $\mathbf{M}_{lr}$. This efficiently preserves the fundamental spatial structure and layout from the original low-resolution image, providing a stable geometric foundation for subsequent detail refinement. For $\mathbf{M}_{HS}$, it is first upscaled via subpixel convolution and then processed through a convolutional layer to obtain $\mathbf{M}_{hc}$. Finally, $\mathbf{M}_{hc}$ and $\mathbf{M}_{SR-bil}$ are fed into the HCCA module to obtain the super-resolved image. This design ensures a synergistic reconstruction, where the structural fidelity preserved by the low-level features and the rich textual details provided by the high-level features complement each other. Through adaptive, channelwise fusion within the HCCA module, both global layout integrity and local fine-grained quality are enhanced, resulting in an output of superior visual fidelity.

IV. Experiments

This section conducts a series of experiments to verify the performance of our HiFi-Net. Section IV-A delineates the experimental setups. Section IV-B presents comprehensive quantitative and qualitative comparisons between the proposed HiFi-Net and SOTA methods.

A. Setups

1) Datasets and Configuration: The performance of the proposed HiFi-Net is assessed on three widely used RSI datasets: UCMerced [49], RSSCN7 [50], and AID [51]. In each dataset, the original images are used as HR ground truth. The corresponding LR images for experimentation are generated by downsampling the HR images via bicubic interpolation. Furthermore, NWPU-RESISC45 [52] is employed to evaluate the performance of HiFi-Net for SR tasks with realistic textures and structures.

1) The UCMerced dataset, introduced by the University of California, is a classic remote sensing scene classification dataset designed for categorizing land-use scenes in urban areas. With a spatial resolution of approximately 0.3 m, each image measures 256×256 pixels. The dataset comprises 21 scene categories, each containing 100 images, resulting in a total of 2100 images. Following the data partitioning strategy used in [19] and [27], UCMerced dataset is divided into three subsets: 945 images for training, 105 for validation, and 1050 for testing.
2) The RSSCN7 dataset contains 2800 RSIs covering seven typical scene categories. Each category includes 400 images sampled at four different scales, each image has a size of 400×400 pixels. Consistent with the partitioning approach applied to the UCMerced dataset, RSSCN7 is split into 1120 training images, 280 validation images, and 1400 testing images.
3) The AID dataset is a large-scale benchmark dataset for aerial scene classification. The dataset consists of 10 000 labeled scene images collected from Google Earth, covering 30 scene categories with approximately 200–400 images per category. Each image has a size of 600×600 pixels, and spatial resolutions ranging from 0.3 to 8 m/pixel. The dataset was partitioned into 7850 images for training, 150 for validation, and 2000 for testing.
4) NWPU-RESISC45 dataset is a publicly available benchmark dataset for RSI scene classification. It contains a

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total of 31 500 images, each measuring 256 × 256 pixels, covering 45 distinct scene classes with 700 images per class. The dataset is characterized by its large scale in both the number of classes and total images, and it exhibits substantial variations in translation, spatial resolution, viewpoint, object pose, illumination, background, and occlusion.

2) Implementation Details: To evaluate the reconstruction capability of the proposed HiFi-Net, we perform experiments at three different upscaling factors: ×2, ×3, and ×4. Correspondingly, the upsampling path in the IRP is adaptively configured based on the target output scale. Specifically, for ×2 and ×3 SR, a single-stage upsampling path is employed, consisting of a convolution layer followed by a pixel-shuffle operation. For the more challenging ×4 SR, a two-stage cascade is employed: a convolution layer and a pixel-shuffle layer are sequentially applied twice to progressively restore HR details. During training, 48 × 48 patches are randomly cropped from LR training images. Corresponding HR patches are cropped from the GT images based on the specific scale factor (e.g., 192 × 192 for ×4 scale). Furthermore, to enhance the robustness of HiFi-Net, extensive data augmentations including random horizontal flipping and rotations by 90°, 180°, and 270° are applied.

The proposed HiFi-Net is optimized using the Adam optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\epsilon = 10^{-8}$. Here, β_1 and β_2 control the exponential decay rates of the first- and second-moment estimates, respectively, while ϵ is a small constant added to maintain numerical stability during the update steps. Each model is trained for 500 epochs with an initial learning rate of $\alpha_0 = 10^{-4}$. The learning rate α is equal to 5×10^{-5} after the 400 epochs. The network is structured with $N = 16$ FFAs, where the channel dimension of the input feature maps for each layer is 64. Inside each FFA module, there are 8 DBs with a growth rate of $D = 64$. All experiments are conducted on the PyTorch 2.4.0 framework with a NVIDIA GeForce RTX 5080 GPU.

3) Evaluation Metrics: The performance of the proposed HiFi-Net is evaluated using four quantitative metrics: peak signal-to-noise ratio (PSNR) [53], structural similarity index (SSIM) [54], spectral angle mapper (SAM) [55], and spatial correlation coefficient (SCC) [56]. PSNR serves as an objective measure of pixelwise fidelity between the GT and the super-resolved image, where a higher value indicates better-reconstructed image quality. SSIM provides a perceptual assessment of image quality by comparing the luminance, contrast, and structural information. It produces a value between −1 and 1, where a value closer to 1 signifies superior structural preservation. SAM is a critical metric for evaluating spectral fidelity. It models the spectrum of each pixel as a high-dimensional vector and measures the spectral similarity by the angle between corresponding vectors. A smaller angle indicates less spectral distortion and a higher probability that the pixels represent the same material. SCC evaluates the degree of spatial pattern alignment, ranging from −1 to 1, and a higher SCC value denotes stronger spatial consistency and better retention of textual details between the compared images.

B. Quantitative and Qualitative Comparisons

We compare the proposed HiFi-Net with thirteen SR methods, including four widely recognized natural image SR methods—SRCNN [20], FSRCNN [35], VDSR [36], and OmniSR [44]—and nine RSI SR-specific methods: LGCNet [29], DCM [30], HSENet [19], FeNet [31], HATNet [43], TransENet [27], HAUNet [32], SRDSRAN [14], and PLAN [48].

1) Quantitative Results: The quantitative comparisons between the proposed HiFi-Net and other thirteen competitive methods are summarized in Table I, HiFi-Net achieves optimal overall performance across three datasets (UCMerced, RSSCN7, and AID) and all upscaling factors (×2, ×3, and ×4). It demonstrates comprehensive superiority over all compared methods in the four evaluation metrics, validating its exceptional SR reconstruction capability across diverse scenes.

For a scale factor of ×2, HiFi-Net attains the best scores across all metrics. Taking PSNR as an example, HiFi-Net achieves 34.54, 30.06, and 35.49 dB on the UCMerced, RSSCN7, and AID datasets, respectively. These results surpass those of the second-best method HAUNet by margins of 0.23, 0.05, and 0.12 dB. Compared to the recently proposed PLAN, HiFi-Net's PSNR (34.54 dB) is significantly higher than that of PLAN (32.12 dB), with a margin of 2.42 dB on the UCMerced dataset. HiFi-Net also leads comprehensively in SSIM, SAM, and SCC. On the UCMerced dataset, SSIM metric of HiFi-Net (0.9339) represents a 0.51% improvement over the second-best method DCM (0.9292). Notably, it achieves the most significant reduction in the SAM metric, which is 3.10% lower than the second-best method OmniSR. On the AID dataset, the SCC of our method exceeds that of the second-best HATNet method by 1.68%.

For a scale factor of ×3, HiFi-Net maintains superior performance across all three datasets. In terms of the AID dataset, HiFi-Net achieves a PSNR of 31.62 dB, which is 0.48% higher than that of the second-best method HSENet (31.47 dB), SAM metric of HiFi-Net is 1.72% lower than that of the second-best method HSENet. Furthermore, on the UCMerced dataset, the PSNR of HiFi-Net (30.28 dB) surpasses that of the second-best method HAUNet by 0.17 dB (a relative improvement of 0.56%), while its SAM shows a reduction of 0.44%. On the RSSCN7 dataset, HiFi-Net achieves a 0.45% higher SCC and a 0.39% lower SAM compared to the second-best method HAUNet.

At the more challenging ×4 scale, HiFi-Net continues to deliver comprehensively leading performance. For instance, in terms of the SAM metric, HiFi-Net outperforms the second-best method on each dataset: it surpasses HATNet by 1.18% on the UCMerced dataset, exceeds HATNet by 0.27% on the RSSCN7 dataset, and leads HAUNet by 1.92% on the AID dataset. Notably, on the texturally complex AID dataset, HiFi-Net establishes a substantial lead in SCC (0.2956), outperforming the second-best method HSENet by a significant margin of 5.31%, further widening the performance gap at higher upscaling factors. Although PLAN achieves 0.44 dB higher PSNR at this scale, HiFi-Net remains more robust across other structural and spectral metrics like

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TABLE I

COMPARISON OF THE PROPOSED HIFI-NET WITH OTHER COMPETITIVE SOTA METHODS ON THREE DATASETS ACROSS ×2, ×3, AND ×4 SCALE, THE BEST AND SECOND-BEST PERFORMANCE METRICS ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Scales	Dataset	Metric	FSRCNN [35]	SRCNN [20]	VDSR [36]	OmniSR [44]	DCM [30]	FeNet [31]	HATNet [43]	HAUNet [32]	HSENet [19]	LGCNet [29]	SRDSRAN [14]	TransENet [27]	PLAN [48]	HiFi-Net (Ours)
×2	UCMerced	PSNR(dB)	31.73	32.72	33.73	32.12	33.92	33.54	32.67	34.31	34.19	33.32	32.72	33.71	33.87	**34.54**
		SSIM	0.9002	0.9144	0.9262	0.9275	0.9292	0.9251	0.9106	0.9291	0.9284	0.9236	0.9104	0.9265	0.8807	**0.9339**
		SAM	0.06431	0.05886	0.05245	0.05366	0.05128	0.05372	0.05018	0.05018	0.05016	0.05457	0.06166	0.05252	0.05015	**0.04864**
		SCC	0.5396	0.5723	0.6141	0.6289	0.6281	0.6118	0.5863	0.6379	0.6381	0.6022	0.6247	0.6148	0.6368	**0.6451**
	RSSCN7	PSNR(dB)	29.64	29.67	29.87	29.73	29.91	29.81	29.11	30.01	29.96	29.68	28.86	29.91	29.97	**30.06**
		SSIM	0.7902	0.7973	0.7977	0.7974	0.7992	0.7948	0.8029	0.8026	0.8022	0.7934	0.7960	0.7990	0.8017	**0.8041**
		SAM	0.1062	0.1043	0.1039	0.1032	0.1031	0.1090	0.1014	0.1016	0.1020	0.1048	0.1160	0.1036	0.1016	**0.1011**
		SCC	0.2601	0.2791	0.2797	0.2796	0.2831	0.2747	0.2207	0.2870	0.2855	0.2731	0.2463	0.2794	0.2561	**0.2902**
	AID	PSNR(dB)	34.51	34.51	35.12	34.87	35.23	35.15	30.12	35.37	35.33	34.77	/	35.14	34.81	**35.49**
		SSIM	0.9287	0.9277	0.9358	0.9371	0.9364	0.9358	0.9125	0.9384	0.9379	0.9316	/	0.9357	0.9292	**0.9398**
		SAM	0.05794	0.05774	0.05392	0.05281	0.05301	0.05358	0.08988	0.05301	0.05245	0.05595	/	0.05348	0.05316	**0.05162**
		SCC	0.6048	0.6016	0.6217	0.6516	0.6345	0.628	0.6559	0.6558	0.6552	0.6103	/	0.6311	0.6533	**0.6669**
×3	UCMerced	PSNR(dB)	28.04	28.76	29.36	28.87	29.69	29.33	28.94	30.11	30.01	29.21	28.91	29.61	29.68	**30.28**
		SSIM	0.7917	0.8088	0.8322	0.7902	0.8370	0.8275	0.8351	0.8439	0.8432	0.8224	0.8391	0.8341	0.8406	**0.8448**
		SAM	0.09749	0.09123	0.08388	0.08252	0.08291	0.08593	0.08026	0.08027	0.08046	0.08747	0.08732	0.08343	0.08056	**0.07991**
		SCC	0.2942	0.3335	0.3886	0.3801	0.3967	0.3703	0.3425	0.4165	0.4108	0.3631	0.4018	0.3824	0.4113	**0.4176**
	RSSCN7	PSNR(dB)	27.65	27.82	27.84	27.88	27.89	27.82	27.98	27.97	27.87	27.72	27.23	27.90	27.97	**28.04**
		SSIM	0.6896	0.6976	0.6979	0.7024	0.7002	0.6984	0.7026	0.7009	0.7044	0.6948	0.6998	0.7005	0.7041	**0.7058**
		SAM	0.134	0.1322	0.1318	0.1311	0.1308	0.1319	0.1303	0.1296	0.1299	0.1325	0.1403	0.1308	0.1298	**0.1291**
		SCC	0.127	0.1477	0.1481	0.1487	0.1501	0.1442	0.0979	0.1571	0.1557	0.1401	0.1301	0.1462	0.1576	**0.1578**
	AID	PSNR	30.65	30.57	31.18	31.58	31.29	31.24	26.91	31.45	31.47	30.81	30.80	31.27	31.43	**31.62**
		SSIM	0.8395	0.8384	0.8518	0.8564	0.8558	0.8541	0.8188	0.8594	0.8591	0.8426	0.859	0.8553	0.8564	**0.8599**
		SAM	0.08867	0.08893	0.08329	0.08133	0.08179	0.08264	0.1302	0.08035	0.08029	0.08688	0.08567	0.08196	0.08042	**0.0789**
		SCC	0.3537	0.3552	0.3829	0.4068	0.3992	0.3953	0.4056	0.4087	0.4084	0.3644	0.3944	0.3975	0.4074	**0.4095**
×4	UCMerced	PSNR(dB)	26.26	26.68	27.39	26.59	27.45	27.27	26.74	27.75	27.66	27.02	26.78	27.47	27.38	**27.88**
		SSIM	0.7052	0.7218	0.7476	0.7156	0.7538	0.7433	0.7484	0.7611	0.7601	0.7347	0.7490	0.7538	0.7522	**0.7631**
		SAM	0.1198	0.1156	0.1072	0.1065	0.1067	0.1086	0.1029	0.1032	0.1048	0.1123	0.1076	0.1062	0.1031	**0.1017**
		SCC	0.1614	0.2041	0.2512	0.2645	0.2587	0.2369	0.2312	0.2763	0.2715	0.2215	0.2728	0.2533	0.2738	**0.2773**
	RSSCN7	PSNR(dB)	26.52	26.65	26.67	26.59	26.68	26.61	26.67	26.79	26.73	26.57	26.15	26.74	26.57	**26.86**
		SSIM	0.6229	0.6212	0.6303	0.6348	0.6329	0.6297	0.6345	0.6358	0.6359	0.6259	0.6274	0.6340	0.6358	**0.6385**
		SAM	0.153	0.1511	0.1507	0.1493	0.1495	0.1511	0.1488	0.1489	0.1496	0.1521	0.1591	0.1499	0.1490	**0.1484**
		SCC	0.06294	0.07696	0.07699	0.07998	s0.07864	0.07421	0.05260	0.08228	0.08191	0.07117	0.0730	0.0789	0.08201	**0.08243**
	AID	PSNR (dB)	28.52	28.47	28.99	**29.97**	29.16	29.05	25.23	29.42	29.31	28.64	28.60	29.22	29.19	29.53
		SSIM	0.7598	0.7567	0.7756	0.7923	0.7822	0.7788	0.7379	0.7923	0.7872	0.7635	0.7922	0.7851	0.7891	**0.7939**
		SAM	0.1111	0.1121	0.1054	0.1011	0.1037	0.1045	0.1583	0.1007	0.1016	0.1140	0.1092	0.1025	0.1065	**0.0988**
		SCC	0.2166	0.2137	0.2440	0.2797	0.2671	0.2659	0.2777	0.2799	0.2781	0.2205	0.2557	0.2728	0.2459	**0.2956**

SSIM and SAM. Specifically, HiFi-Net achieves an SSIM of 0.7939, surpassing the competitive HAUNet by 0.20%. On the RSSCN7 dataset, HiFi-Net achieves a 0.41% higher SSIM than the second-best method HSENet. These results indicate that HiFi-Net can effectively recover details and maintain favorable spectral fidelity under medium upscaling factors.

In summary, the proposed HiFi-Net method achieves superior performance across all datasets and all evaluation metrics at ×2 to ×4 upscaling factor. This robust and comprehensive superiority fully demonstrates the effectiveness and excellence of the proposed HiFi-Net, which innovatively integrates Fourier transformation with hierarchical attention mechanisms for RSI SR.

2) Qualitative Results: In addition, Fig. 4 presents several samples of reconstructed images of HiFi-Net and the competing methods on the UCMerced dataset at different scales. In order to highlight the difference in visual representation between images obtained by the proposed HiFi-Net and other competing methods, we select and magnify two localized regions from the reconstructed images for detailed visualization and comparison.

Compared with other competing methods, HiFi-Net reconstructs images with richer textural details and significantly fewer artifacts. As shown in the "airplane53" at the ×4 scale, DCM and FENet have artifacts and perform poorly on some small details. HSNet, TransNET, and HAUNet seem to better preserve the spatial and spectral details of the image, but the proposed HiFi-Net still gains the best performance. Our method not only achieves the best PSNR and SSIM scores but also recovers sharper and more accurate edges of the airplane without color blending, resulting in an output that is more visually similar to the GT.

Similarly, Fig. 5 compares the SR results of different methods in the image "airport_112" from the AID dataset at ×4 scale. FSRCNN and LGCNet produce ambiguous results that are only slightly better than the SRCNN. Models such as VDSR, DCM, and FENet exhibit varying degrees of blurring, which can be attributed to insufficient feature utilization.

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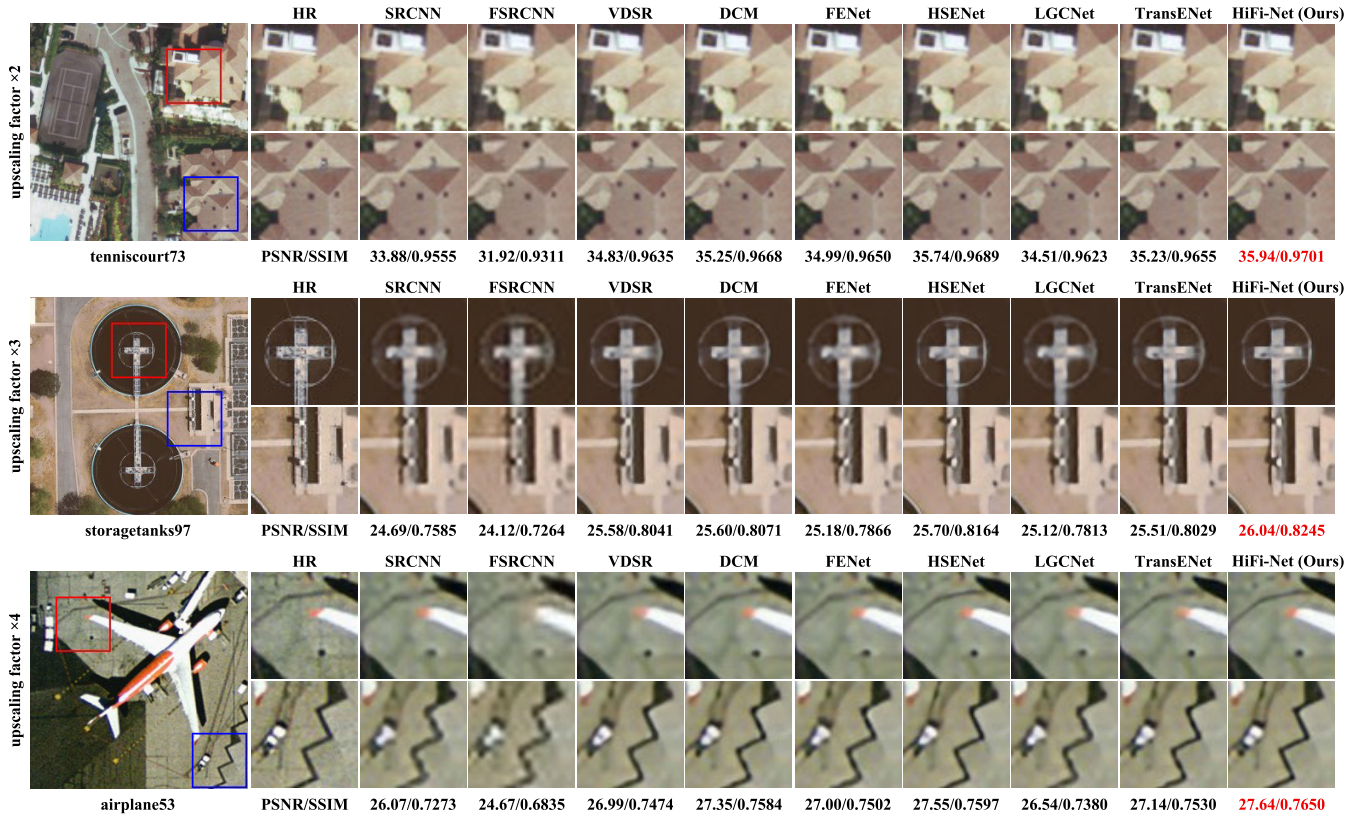


Fig. 4. Comparison of the proposed HiFi-Net with other competitive SOTA methods on the UCMerced dataset under ×2, ×3, and ×4 scale. The red and blue boxes indicate the magnified regions for detailed visual comparison.

While HSNet and HAUNet generate more texture details, their outputs still show a noticeable gap compared to the HR image. In contrast, HiFi-Net effectively leverages multilevel features, which produce clearly defined boundaries between red and gray areas and ultimately lead to superior detail preservation and more accurate shape reconstruction.

The consistent alignment between the quantitative metrics and qualitative observations further underscores the superiority of the proposed HiFi-Net.

V. Discussions

A series of ablation studies are conducted to further investigate the effectiveness of the proposed HiFi-Net. All models are trained and evaluated under the same experimental setup to ensure fair comparison. First, we assess the contributions of the three core customized modules: FFA, HSCA, and HCCA. Subsequently, we investigate the impact of varying the quantity of FFAs within HiFi-Net. Finally, we compare the model complexity of HiFi-Net against SOTA methods.

A. Contribution of Core Modules

To validate the effectiveness of the HiFi-Net architecture and evaluate the contribution of the proposed components, we design a series of scenarios by systematically removing one or several core modules. As detailed in Section III, HiFi-Net comprises three core modules: FFA, HSCA, and HCCA. By selectively ablating these components, we construct seven distinct experimental scenarios: the proposed HiFi-Net without any of the three modules removed, Net-S1–Net-S3 with only one module removed, Net-S4–Net-S6 with two modules removed, and Net-S7 with no module retained. A comparative analysis of these seven scenarios is presented in Table II.

TABLE II
Modulewise Ablation Comparisons of HiFi-Net Components Contributions on the UCMerced Dataset

Modules	FFA	HSCA	HCCA	PSNR(dB)	SSIM	SAM	SCC
HiFi-Net (Default)	✓	✓	✓	**34.54**	**0.9339**	**0.04864**	**0.6451**
Net-S1		✓	✓	34.51	0.9338	0.04871	0.6449
Net-S2	✓		✓	34.47	0.9337	0.04873	0.6450
Net-S3	✓	✓		34.49	0.9335	0.04868	0.6447
Net-S4	✓			33.34	0.9301	0.04908	0.6419
Net-S5		✓		34.18	0.9331	0.04877	0.6435
Net-S6			✓	34.12	0.9332	0.04879	0.6446
Net-S7				33.40	0.9304	0.04905	0.6418

As shown in Table II, the HiFi-Net (default) equipped with all three modules achieves the best performance across all evaluation metrics with 34.54 dB PSNR, 0.9339 SSIM, 0.04864 SAM, and 0.6451 SCC. The removal of any module from HiFi-Net architecture consistently leads to performance degradation. Specifically, removing the FFA module (Net-S1) results in the most pronounced decline in PSNR (0.03 dB

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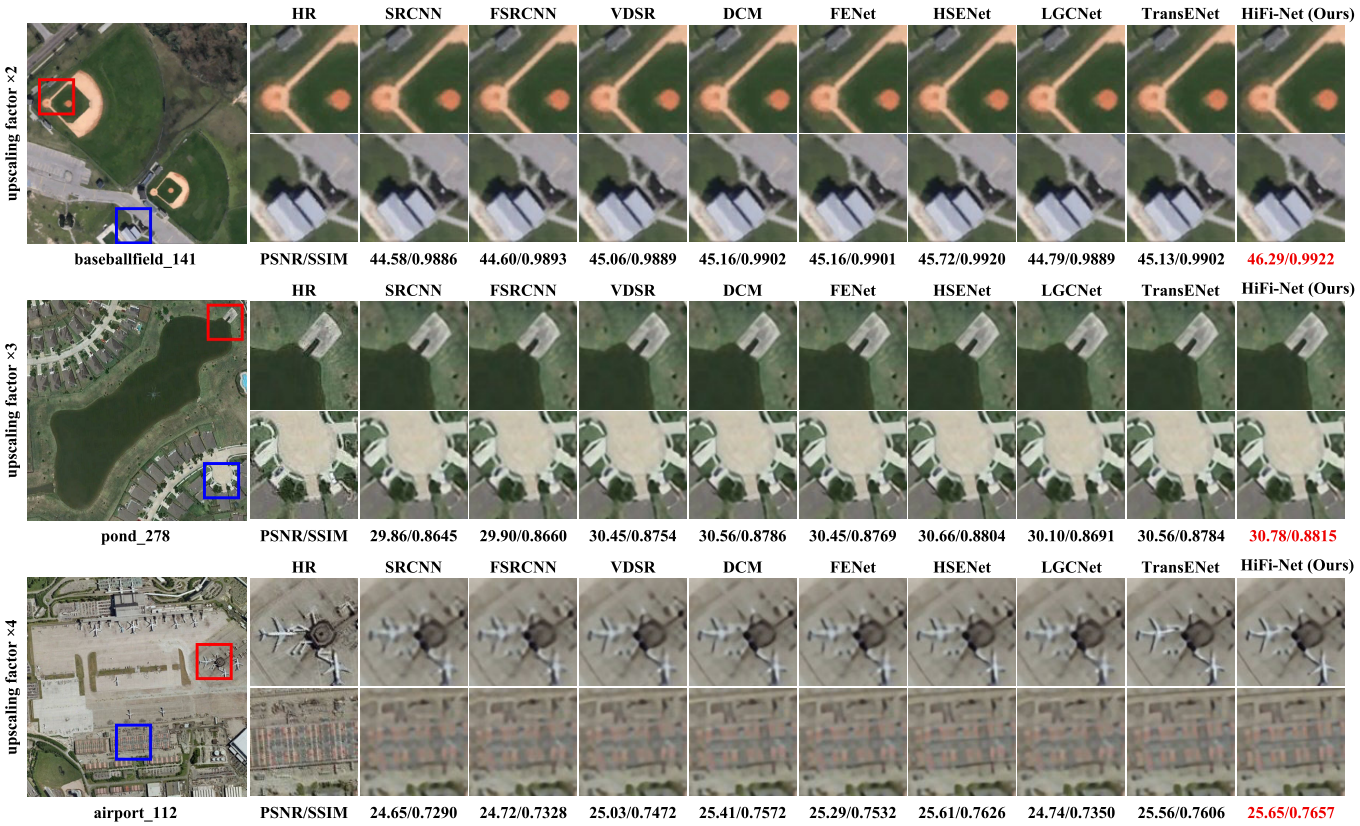


Fig. 5. Comparison of the proposed HiFi-Net with other competitive methods on AID dataset for ×2, ×3, and ×4 scale. The red and blue boxes indicate the magnified regions for detailed visual comparison.

reduction), indicating its crucial role in enhancing pixel-level fidelity. The absence of the HSCA module (Net-S2) affects PSNR decreasing by 0.07 dB (0.19%), SSIM decreasing by approximately 0.02%, SAM increasing by 0.19%, and SCC decreasing b 0.02%. When the HCCA module is removed (Net-S3), we observe the most significant deterioration in SSIM (0.04% reduction) and SAM (0.08% increase), underscoring the importance for structural preservation and spectral fidelity of HCCA.

More profound insights emerge when examining configurations with two modules removed from HiFi-Net. The network retaining only the FFA module (Net-S4) exhibits the most substantial performance drop among all variants, with PSNR decreasing by 1.20 dB and SSIM by 0.0038 compared to the defaulted HiFi-Net. This suggests that while FFA alone provides some baseline capability, it cannot effectively compensate for the loss of spatial and channel attention mechanisms. Configurations preserving both HSCA and HCCA modules but lacking FFA (Net-S5) demonstrate relatively stable performance, particularly in SSIM (0.9331) and SAM (0.04877), highlighting the complementary relationship between spatial and channel hierarchical attention mechanisms.

Overall, the consistent metric variations observed in the ablation study provide compelling evidence that each proposed module contributes uniquely to the overall performance. Specifically, HCCA plays an indispensable role in achieving high-fidelity reconstruction, FFA contributes primarily to spectral fidelity and pixel-level accuracy, and HSCA exerts a notable influence on structural preservation. Furthermore, the core modules operate synergistically rather than independently within HiFi-Net.

B. Effectiveness of Frequency-Enhanced Frequency–Spatial Attention Module

To validate the effectiveness of the proposed FFA module in HiFi-Net, we design seven model variants by setting the number of FFAs to $\{0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6\}$. Net-V1 does not contain the FFA module and instead uses a standard attention module as a replacement, while Net-V2 to Net-V7 progressively increase the number of FFAs in powers of two to examine its impact. Results on the UCMerced dataset across multiple scales (Table III) demonstrate the superior performance of the proposed FFA over the standard attention module. Notably, even with only two FFAs, Net-V2 outperforms Net-V1 by 0.15 dB in PSNR and 0.0008 in SSIM at the ×2 scale. As the number of FFAs increases from 2^1 to 2^6, performance exhibits a nonmonotonic relationship: both PSNR and SSIM generally improve until an optimal point is reached, beyond which they plateau or slightly decline. For instance, at ×2 scale, PSNR increases from 34.32 dB (Net-V2) to 34.61 dB (Net-V7), while SSIM improves from 0.9331 to 0.9357. At ×3 and ×4 scale, the PSNR initially rises to 30.31 and 27.90 dB, then declines to 30.28 and 27.82 dB, respectively.

To provide an intuitive visualization, Fig. 6 illustrates the trends of PSNR and SSIM with varying numbers of FFAs in

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TABLE III
QUANTITATIVE COMPARISON OF SEVEN FFAS VARIANTS (NET-V1–V7) ON THE UCMERCED DATASET ACROSS THREE SCALE FACTORS

Scale factors		×2		×3		×4	
Models	FFAs	PSNR(dB)	SSIM	PSNR(dB)	SSIM	PSNR(dB)	SSIM
Net-V1	0	34.17	0.9323	29.92	0.8324	27.68	0.7589
Net-V2	2^1	34.32	0.9331	30.05	0.8346	27.72	0.7608
Net-V3	2^2	34.41	0.9334	30.13	0.8373	27.75	0.7634
Net-V4	2^3	34.45	0.9337	30.21	0.8407	27.81	0.7656
Net-V5	2^4	34.53	0.9338	30.26	0.8421	27.90	0.7691
Net-V6	2^5	34.58	0.9355	30.31	0.8436	27.88	0.7677
Net-V7	2^6	34.61	0.9357	30.28	0.8431	27.82	0.7656

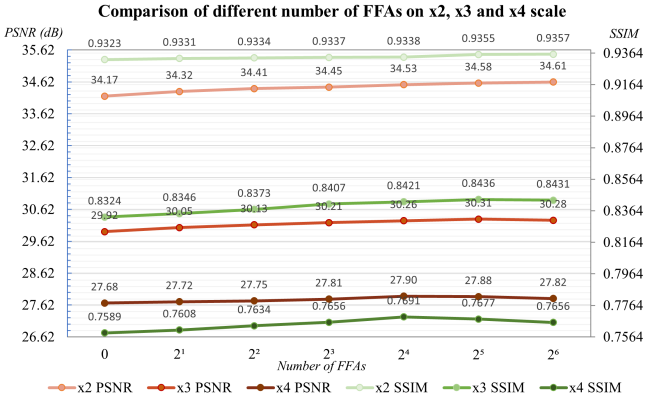


Fig. 6. Performance comparison of different quantities of FFAs on UCMerced dataset under ×2, ×3, and ×4 scale.

HiFi-Net. The x-axis represents the number of FFAs, while the y-axis denotes the corresponding metric values. Red and blue markers correspond to PSNR and SSIM, respectively, with color intensity indicating the scale factor (lightest for ×2, darkest for ×4). It can be observed that the optimal FFA numbers vary across scale factors. At ×2 scale, performance improves as the number of FFAs increases, but exhibits a saturation tendency. At ×3 scale, PSNR peaks at around 2^5 FFAs, whereas SSIM attains the optimal value at around 2^4 FFAs. At ×4 scale, Net-V5 (2^4 FFAs) achieves the best PSNR (27.90 dB) and SSIM (0.7691). This scale-dependent behavior suggests that higher scale factors do not necessarily benefit from more FFAs, instead, a moderate number of FFAs can prevent overparameterization while maintaining competitive performance. The most significant gains occur within the lower range of FFA counts (2^1–2^4), with diminishing returns observed beyond 2^5 FFAs. The subsequent performance drop is primarily attributed to optimization difficulties arising from excessive model complexity, which hinders convergence to the optimal solution. Therefore, balancing model complexity and performance, we ultimately set the number of FFAs in the HiFi-Net to 2^4.

C. Feature Visualization

To more intuitively validate the effectiveness of the proposed module in HiFi-Net, we select images of different scenes from the UCMerced, AID, and RSSCN7 datasets and conducted feature visualization using the trained model at a scale factor of ×4. As shown in Fig. 7, the first column displays the HR images. The second to fourth columns respectively show the visualization results obtained by averaging the feature maps along the channel dimension after processing by the final FFA module, the HSCA module, and the HCCA module. Warmer colors (red/yellow) indicate higher activation magnitudes, while cooler colors (blue) indicate lower activity. The fifth column presents the super-resolved reconstruction results along with the PSNR and SSIM metrics.

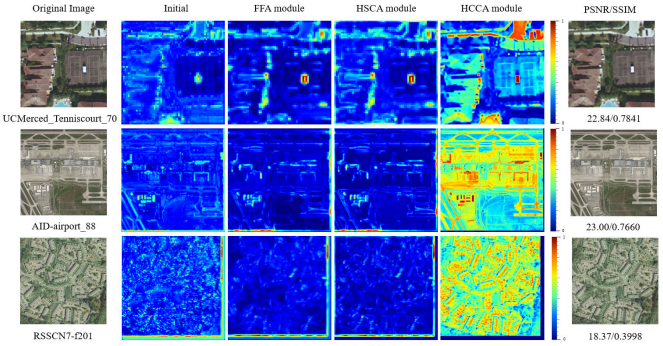


Fig. 7. Visualization of intermediate features from the core modules in HiFi-Net for different images selected from three datasets at ×4 scale.

From the color and brightness distribution of the feature maps, it can be observed that the FFA, HSCA, and HCCA modules exhibit distinct focus regions and enhancement intensities for image features. Specifically, the feature maps from the FFA module are predominantly blue and show diffuse responses, indicating that the FFA module tends to capture the overall scene structure. The HSCA module also focuses mainly on blue regions, but its responses are more concentrated, suggesting that it strengthens the contours of primary objects and internal structural textures, demonstrating its capability in modeling spatial relationships. In contrast, the HCCA module pays greater attention to warmer color regions, visually highlighting its role in enhancing high-frequency details and texture information. The three modules operate synergistically, collectively improving the SR reconstruction performance of HiFi-Net.

D. Effectiveness in Real-World Scenarios

We use images from the NWPU-RESISC45 dataset [52] without any simulated degradation to validate the effectiveness of the proposed HiFi-Net in real-world scenarios. Three representative scenes such as airport, tennis court, and church are selected for evaluation. Two no-reference perceptual metrics—the natural image quality evaluator (NIQE) and the blind/referenceless image spatial quality evaluator (BRISQUE)—are employed to evaluate the quality of the reconstructed images, where lower NIQE and BRISQUE scores indicate better visual quality and closer adherence to natural image statistics.

As shown in Fig. 8, the first column presents the original high-resolution images, columns two to five display the SR

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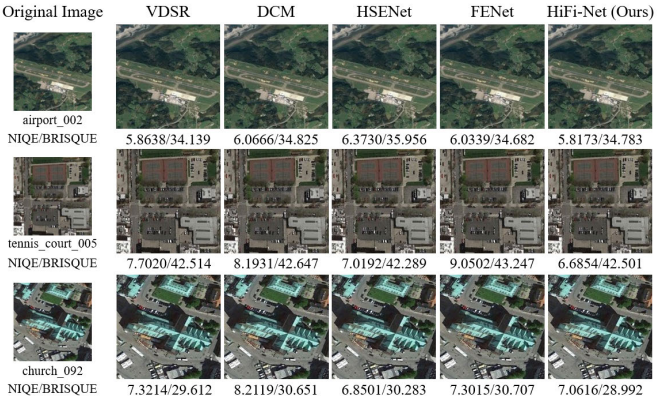


Fig. 8. Comparison of the proposed HiFi-Net and the competing methods of visual and perceptual indicators (NIQE and BRISQUE) on the NWPU-RESISC45 dataset at $\times 2$ scale.

TABLE IV

COMPUTATIONAL EFFICIENCY COMPARISON OF HIFI-NET AGAINST SEVERAL SOTA SR METHODS ON UCMERCED DATASET AT $\times 4$ SCALE

Methods	PSNR(dB)	SSIM	Params (M)	FLOPs (G)	Inference Time (ms)
HiFi-Net (Ours)	**27.88**	**0.7631**	26.01	101.34	31.51
VDSR	27.39	0.7476	0.6713	44.03	13.56
DCM	27.45	0.7538	2.1724	13.01	3.31
FeNet	27.27	0.7433	0.36618	1.443	28.54
HSENet	27.66	0.7601	5.433	19.20	34.27
LGCNet	27.02	0.7347	3.998	6.464	37.75
TransENet	27.47	0.7538	37.46	21.44	32.23
HAUNet	27.75	0.7611	2.632	6.179	53.69
HATNet	26.74	0.7484	20.363	86.408	211.512
SRDSRAN	26.78	0.7490	11.070	47.762	78.209

results from different comparative methods, and the last column shows the output of our proposed HiFi-Net. Visually, it is evident that HiFi-Net produces clearer and more natural details and textures, resulting in reconstructions that are closer to the real scenes. The corresponding perceptual metrics further support this observation. Taking the tennis court scene as an example, compared to FENet, HiFi-Net reduces the NIQE score by approximately 35.4% (from 9.0502 to 6.6854) and lowers the BRISQUE score by about 1.75% (from 43.247 to 42.501). These results confirm the effectiveness of HiFi-Net and superiority in real-world RSI SR tasks.

E. Computational Efficiency Evaluation

In order to evaluate the computational efficiency of our proposed HiFi-Net, we compare it against SOTA SR methods, including VDSR, DCM, LGCNet, FeNet, HSENet, TransENet, HAUNet, HATNet, and SRDSRAN, using the UCMerced dataset at the $\times 4$ scale. As summarized in Table IV, HiFi-Net contains 26.01 M parameters and requires 101.34 G FLOPs, indicating a higher computational capacity while achieving the best reconstruction quality (27.88 dB PSNR and 0.7631 SSIM) among the compared methods. Although its FLOPs are higher than those of HAUNet and SRDSRAN, this trade-off is justified by the notable performance gains. Specifically, compared with the HAUNet, which attains the second-best reconstruction metrics, our HiFi-Net demonstrates competitive computational efficiency, with an inference time of 31.51 ms, faster than HAUNet's 53.69 ms. In addition, HiFi-Net features a low memory footprint of 352 MB, whereas HAUNet and TransENet require 2522 and 734 MB, respectively. A more pronounced advantage is observed over HSENet (34.27 ms) and TransENet (32.23 ms). This efficiency can be attributed to our optimized network architecture that balances computational load across different components. While the FFA module in HiFi-Net introduces additional computational cost, this overhead is justified by substantial restoration quality gains essential for remote sensing, where image fidelity directly impacts downstream tasks. Furthermore, the frequency-domain attention mechanism via FFT offers greater efficiency and suitability for resource-constrained scenarios than spatial attention. In comparison, although DCM attains relatively low FLOPs (13.01 G) and fast inference (3.31 ms), it suffers from inferior reconstruction quality (27.45 dB PSNR). Overall, HiFi-Net can rapidly generate superior SR results, offering an excellent trade-off between restoration performance and practical inference efficiency.

VI. CONCLUSION

This article presents HiFi-Net, a frequency–spatial dual-domain attention network for RSI SR with notable architectural innovations. HiFi-Net innovatively integrates frequency-domain global modeling with spatial-domain cross-level modulation, enabling it to focus on global structural features while preserving local details. Specifically, the FAB component in the FFA module transforms features into the frequency domain, boosting global feature extraction capacity to capture long-range dependencies and cross-scale structural patterns, thus overcoming the locality limitation of conventional convolutions. Additionally, HiFi-Net incorporates a hierarchical interactive attention mechanism via the designed HSCA and HCCA modules, which effectively fuses high- and low-level critical features, significantly improving texture recovery authenticity and structural consistency.

Comprehensive evaluation on several benchmark remote sensing datasets demonstrates that HiFi-Net achieves optimal performance across several key metrics. Its exceptional generalization capability is thoroughly validated across diverse scenes and varying scale factors. Systematic ablation studies further corroborate the effectiveness of each core module. Moreover, model efficiency analysis reveals that HiFi-Net leverages the inherent computational simplicity of operations in the frequency domain, maintaining competitive inference speed while delivering superior reconstruction quality, successfully achieving an optimal balance between performance and efficiency.

Although HiFi-Net has achieved promising performance, there is still room for improvement in its inference efficiency. For instance, the fixed number of FFA modules may not be optimal for all image scales and content types. Moreover, despite its efficiency, the current model may still incur considerable computational overhead for deployment on

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lightweight platforms such as microsatellites, where computational resources are highly constrained. Therefore, future work could explore several promising directions, including real-world degradation modeling to better simulate practical acquisition conditions, improving cross-sensor and cross-region generalization through domain adaptation or invariant feature learning, investigating model compression techniques to enable onboard processing, designing adaptive architectures that can dynamically adjust their depth and attention mechanisms, and adopting task-driven evaluation metrics that directly measure the impact of SR on downstream tasks such as land cover classification. Addressing these aspects will further improve the applicability and robustness of HiFi-Net in real-world remote sensing scenarios.

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